

Empirical Methods for Policy Evaluation

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Part 2: Regression Discontinuity Designs and School Choice Models

① RD methods

- ⇒ Local randomization approach
- ⇒ Continuity-based approach
- ⇒ Extensions

② School choice models

- ⇒ Agarwal-Somaini (ARE, 2020)

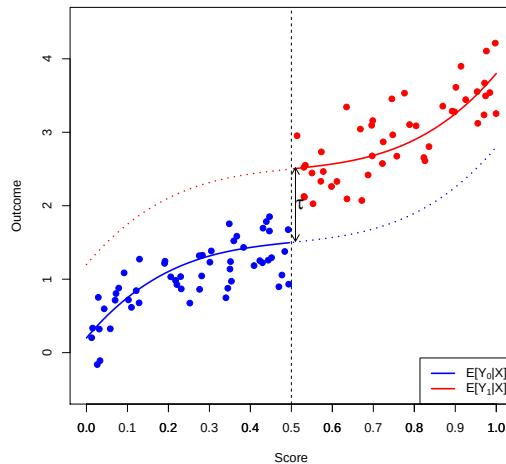
③ Teacher sorting and student outcomes

- ⇒ Bobba-Ederer-Leon-Nielson-Nieddu (JPE, forthcoming)

Setup and Notation

- Potential outcomes: $Y_i(1), Y_i(0)$, with $\tau_i = Y_i(1) - Y_i(0)$
- Running variable (score): X_i
- Treatment indicator: $D_i = D_i(X_i) = 1$ if treated, 0 otherwise
- Observed outcome: $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$
- RD exploits a **discontinuity in $P[D_i = 1|X_i]$** at some cutoff c

Graphical Intuition



The local randomization approach

RD as a Randomized Experiment

- Close enough to the cutoff, some units were “lucky”
- Treatment **as if randomly assigned** in window $W_0 \in [c - w, c + w]$ for $w > 0$:
 ⇒ Probability distribution of X_i is uncounfounded

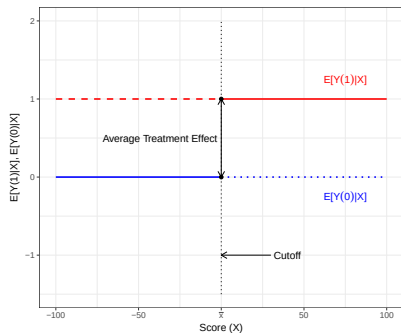
$$P[X_i \leq x | y_i(0), y_i(1)] = P[X_i \leq x], \quad \forall X_i \in W_0$$

- ⇒ Potential outcomes not affected by value of the score

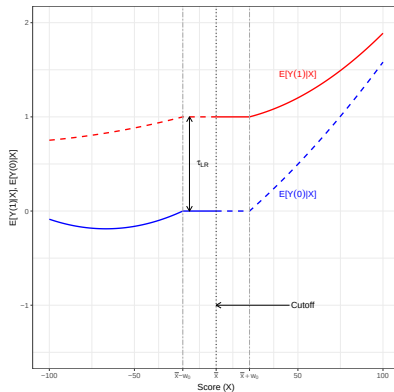
$$Y_i(d, x) = Y_i(d)$$

- Example: $X_i \sim U[0, 1]$, $D_i = \mathbb{I}(X_i \geq c)$, then $P[D_i = 1] = 1 - c$

RD as a Randomized Experiment



(a) Randomized Experiment



(b) RD Design

Window Selection

- Under random assignment, **covariates should be balanced**

$$P[V_i \leq v | D_i = 1] = P[V_i \leq v | D_i = 0]$$

- Can use this idea as a **windows selection criterion**

⇒ Find window in which all covariates are balanced

- Heuristic procedure:

⇒ Choose a test statistic (diff. means, Kolmogorov-Smirnov,...)

⇒ Choose an **initial "small" window** $W_0^{(1)} = [c - w_{(1)}, c + w_{(1)}]$

⇒ Test $H_0 : E[V_i | X_i = c + w_{(1)}] = E[V_i | X_i = c - w_{(1)}]$

⇒ Pick a new, enlarged $W_0^{(k)}$ and **repeat for $k > 1$** until H_0 is rejected

Estimation and inference

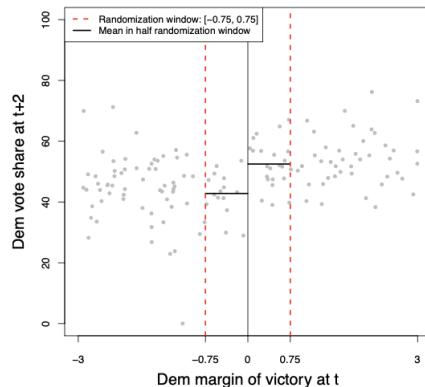
- Once W_0 is found, proceed as in a randomized experiment

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$$

- Covariate-balance criterion may yield windows with few observations
- Inference based on large-sample approximations may not be reliable
- Alternative (and better!) approach: [randomization inference](#)

Example: Incumbency Advantage in U.S. Senate

- Y_i = election outcome at $t + 2$ (= 1 if party wins)
- D_i = election outcome at t (= 1 if party wins)
- X_i = margin of victory at t ($c = 0$)



The Continuity-based approach

Identification

- **Sharp design:** $D_i = \mathbb{I}(X_i \geq c)$
- **Smoothness:** $\mathbb{E}[Y_i(0)|X_i = x]$, $\mathbb{E}[Y_i(1)|X_i = x]$ continuous at $x = c$

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0)|X_i = c]$$

- Treatment effect is nonparametrically identified **only at the cutoff**
 - $\Rightarrow$ We have zero observations at $X_i = c$, so need to approximate the regression functions
 - $\Rightarrow$ Approximation error may directly affect estimation and inference

Identification

- Naive difference in means:

$$\begin{aligned}\Delta(h) &= \mathbb{E}\{Y_i \mid X_i \in [c, c+h]\} - \mathbb{E}\{Y_i \mid X_i \in [c-h, c]\} \\ &= \mathbb{E}\{Y_i(1) \mid X_i \in [c, c+h]\} - \mathbb{E}\{Y_i(0) \mid X_i \in [c-h, c]\} \\ &= \mathbb{E}\{\tau_i \mid X_i \in [c, c+h]\} + \text{Bias}(h)\end{aligned}$$

where $\text{Bias}(h) = E\{Y_i(0) \mid X_i \in [c, c+h]\} - E\{Y_i(0) \mid X_i \in [c-h, c]\}$

- If $\mathbb{E}[Y_i(d) \mid X_i = x]$ is **continuous at $x = c$** for $d = 0, 1$, then:

$$\lim_{h \downarrow 0} \Delta(h) = \mathbb{E}[\tau_i \mid X_i = c]$$

Estimation

- Global Parametric

- ⇒ Estimate a polynomial on full sample
- ⇒ Sensitive to misspecification
- ⇒ Erratic behavior at boundary points

- Local parametric

- ⇒ Estimate a polynomial within an ad-hoc bandwidth
- ⇒ Sensitive to misspecification and bandwidth choice

- Local non-parametric

- ⇒ Data-driven bandwidth selection
- ⇒ Accounts for misspecification when performing inference

Global Parametric Approach

- Parametric assumption on conditional expectations, e.g.

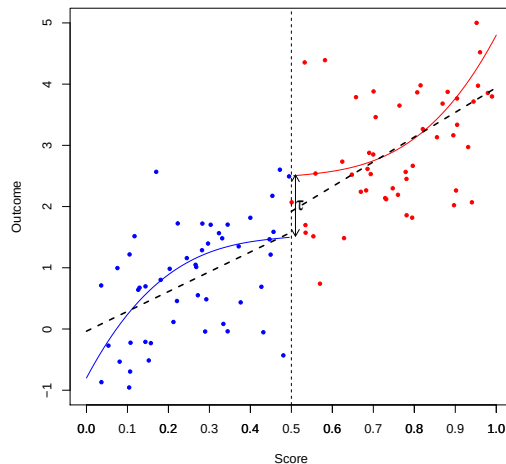
$$\mathbb{E}[Y_i(d)|X_i] = \alpha_d + \beta_d(X_i - c)$$

- This implies

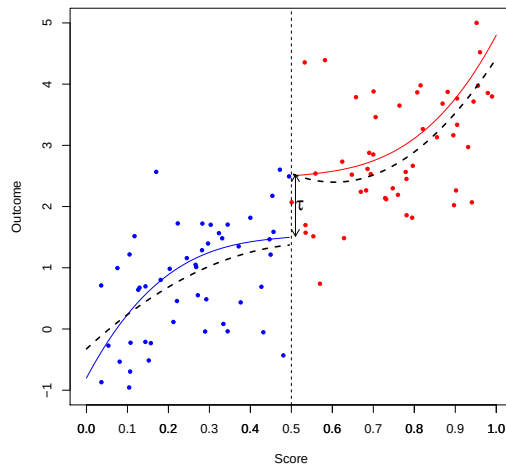
$$Y_i = \alpha_0 + (\alpha_1 - \alpha_0)D_i + \beta_0(X_i - c) + (\beta_1 - \beta_0)(X_i - c)D_i + u_i$$

- Easily estimated by OLS on full sample
- Coefficient $\alpha_1 - \alpha_0$ recovers the treatment effect at the cutoff

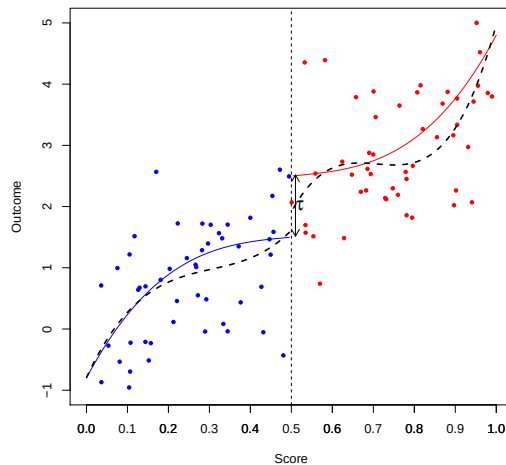
Global Parametric Approach: $p = 1$



Global Parametric Approach: $p = 2$



Global Parametric Approach: $p = 3$



Local Linear (Polynomial) Regression

- Suppose $c = 0$ (otherwise, use $X_i - c$)
- Choose **some bandwidth** $h > 0$ and estimate by OLS:

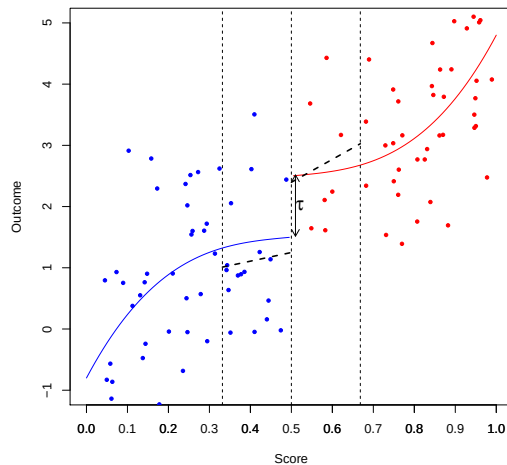
$$(\hat{\alpha}^+, \hat{\beta}^+) = \underset{(\alpha, \beta)}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - \alpha - \beta X_i)^2 \mathbb{I}(0 \leq X_i \leq h)$$

$$(\hat{\alpha}^-, \hat{\beta}^-) = \underset{(\alpha, \beta)}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - \alpha - \beta X_i)^2 \mathbb{I}(-h \leq X_i < 0)$$

- Estimated **treatment effect at the cutoff**

$$\hat{\tau} = \hat{\alpha}^+ - \hat{\alpha}^-$$

Local Linear Regression: Graphical Intuition



Local Linear Regression: OLS Estimands

- By standard OLS algebra (symmetrically for $\hat{\beta}^-, \hat{\alpha}^-$):

$$\hat{\beta}^+ = \frac{\sum_{i=1}^n Y_i (X_i - \bar{X}_h) \mathbb{I}(0 \leq X_i \leq h)}{\sum_{i=1}^n X_i (X_i - \bar{X}_h) \mathbb{I}(0 \leq X_i \leq h)}$$

$$\hat{\alpha}^+ = \bar{Y}_h - \bar{X}_h \hat{\beta}^+$$

- $\bar{X}_h, \bar{Y}_h$ are **local means**

$$\bar{X}_h = \frac{\sum_{i=1}^n X_i \mathbb{I}(0 \leq X_i \leq h)}{\sum_{i=1}^n \mathbb{I}(0 \leq X_i \leq h)}$$

$$\bar{Y}_h = \frac{\sum_{i=1}^n Y_i \mathbb{I}(0 \leq X_i \leq h)}{\sum_{i=1}^n \mathbb{I}(0 \leq X_i \leq h)}$$

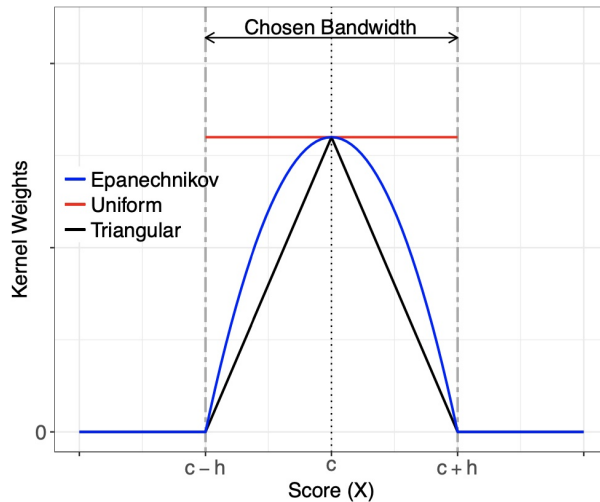
Local Linear Regression: Bias

- If $h = 0$ the estimator would be unbiased, but with $h = 0$ there are no obs. within the bandwidth!

$$\mathbb{E}[\hat{\alpha}^+ | X_i] = \alpha_1 + \beta_1 X_i + h^2 \mathcal{B}_+ + o_p(h^2)$$

- Smaller h implies small bias but fewer observations: more variance
- $\mathcal{B}_+$ is a constant that depends on:
 - ⇒ The curvature of $\mathbb{E}[Y_i(1) | X_i]$ (i.e. approximation error due to higher order terms, $p + 1$)
 - ⇒ The kernel function (i.e. the choice of how we weight observations within the bandwidth)

Different Kernel Weights for RD Estimation



Variance

- Similarly, it can be shown that:

$$\mathbb{V}[\hat{\alpha}^+|\mathbf{X}] = \frac{\mathcal{V}_+}{nh} + o_p(h)$$

- Decreasing the variance requires $nh \rightarrow \infty$ (effective sample size)
- $\mathcal{V}_+$ is a constant that depends on:
 - $\Rightarrow \mathbb{V}[Y_i(1)|X_i = 0]$
 - $\Rightarrow$ The density of the score variable at the cutoff
 - $\Rightarrow$ The kernel function

MSE and Bandwidth Selection

- Mean-squared error (MSE) of the estimated treatment effect at the cutoff

$$\begin{aligned}\text{MSE}(\hat{\tau}) &= \text{Bias}(\hat{\tau})^2 + V[\hat{\tau}] \\ &= h^4 \mathcal{B}^2 + \frac{\mathcal{V}}{nh}\end{aligned}$$

- Trade-off in bandwidth selection: bias vs variance
- MSE-optimal bandwidth:

$$\begin{aligned}h_{\text{MSE}}^* &= \underset{h}{\operatorname{argmin}} \text{MSE}(\hat{\tau}) \\ &= \left(\frac{\mathcal{V}}{4\mathcal{B}^2} \right)^{1/5} n^{-1/5}\end{aligned}$$

Inference

- Under regularity and rate restrictions on $h \rightarrow 0$

$$\sqrt{nh}(\hat{\tau} - \tau - h^2\mathcal{B}) \rightarrow_{\mathcal{D}} \mathcal{N}(0, \Omega)$$

- Where $\Omega \propto \frac{\nu_+ + \nu_-}{f}$ can be estimated using $\epsilon_{+,i} = Y_i - \alpha_1 - \beta_1 X_i$, $\epsilon_{-,i} = Y_i - \alpha_0 - \beta_0 X_i$
- MSE-optimal bandwidth is “too large” to remove the leading bias term $\mathcal{B}$. Two main approaches:
 - $\Rightarrow$ **Undersmoothing**: use a “smaller” bandwidth
 - $\Rightarrow$ **Bias correction**

Undersmoothing

- Choose a small enough bandwidth and proceed as if $\mathcal{B} \approx 0$

$$\sqrt{nh}(\hat{\tau} - \tau - h^2\mathcal{B}) = \sqrt{nh}(\hat{\tau} - \tau) + o_p(1) \rightarrow_{\mathcal{D}} \mathcal{N}(0, \Omega)$$

- Recall that $h_{\text{MSE}}^* \propto n^{-1/5}$ so this approximation will have a bias
- No transparent criteria for shrinking the bandwidth below the MSE-optimal value

Bias Correction

- Estimate **bias term** $\hat{\mathcal{B}}_n$ with pilot bandwidth b
- Standard bias correction ($h/b \rightarrow 0$):

$$\sqrt{nh}(\hat{\tau} - \tau - \mathcal{B}_n) = \sqrt{nh}(\hat{\tau} - \mathbb{E}[\hat{\tau}|\mathbf{X}]) \rightarrow_{\mathcal{D}} \mathcal{N}(0, \Omega)$$

- **Robust bias correction**

$$\sqrt{nh}(\hat{\tau} - \tau - \hat{\mathcal{B}}_n) \rightarrow_{\mathcal{D}} \mathcal{N}(0, \Omega + \Sigma)$$

- Σ accounts for the estimation of the bias

Validation and Falsification of RDD: Density Discontinuity Tests

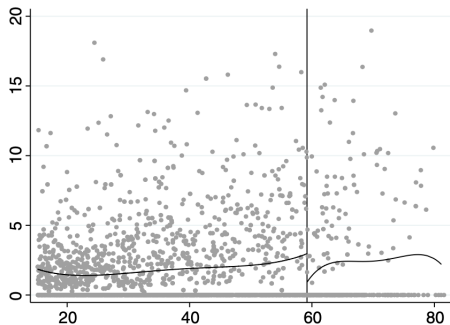
- RDDs can be invalid if individuals manipulate X_i
- Manipulation can imply sorting on one side of the cutoff
- Test whether the density of X_i is continuous around c
- McCrary (2008), Cattaneo, Jansson and Ma (2018)

Validation and Falsification of RDD: Continuity in Covariates

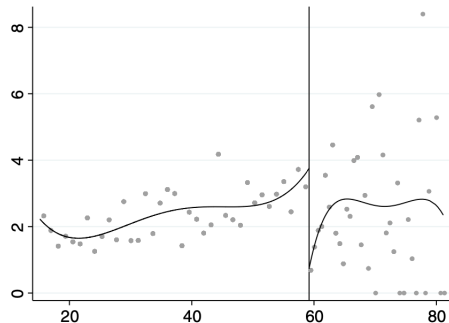
- Some variables should reveal no treatment effect:
 - ⇒ Outcomes not targeted by treatment (**placebo outcomes**)
 - ⇒ Exogenous or **predetermined covariates**
 - Estimate an RD effect on these variables
- ⇒ Finding a **non-zero effect suggests an invalid RDD**
- Existence of other (unobserved) treatments at the cutoff
 - Selection

Example: Impact of Head Start on Child Mortality

- Federal program that provides health and social services for children aged 5-9
 - ⇒ HS assistance for 300 counties based on poverty index ($X_i \geq 59.19$)
 - ⇒ Y_i = county-level mortality rates per 100,000

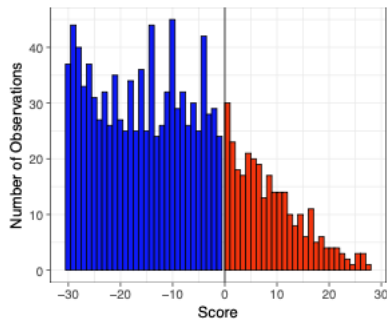


(a) Scatter Plot, Raw Data, $N^- = 2,455$, $N^+ = 290$

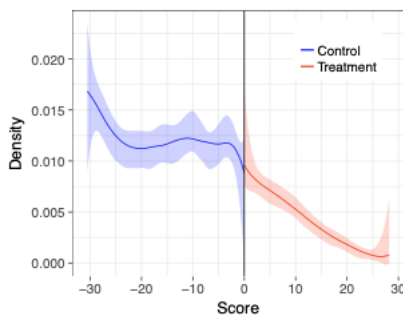


(b) RD Plot, ES, and MV, $J_- = 37$, $J_+ = 38$

Impact of Head Start on Child Mortality: Density of the Poverty Index



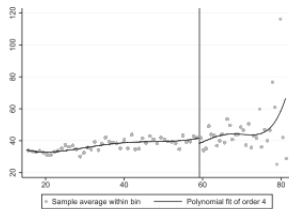
(a) Histogram



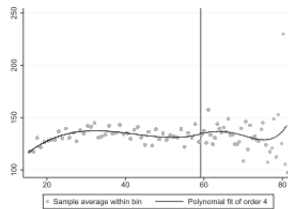
(b) Estimated Density

- Local polynomial density estimator that leads to a hypothesis test based on RBC inference
- T-statistic= -1.394 , p -value= 0.1633

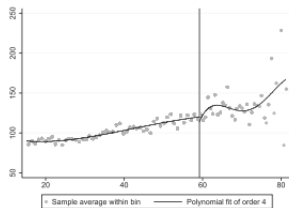
Impact of Head Start on Child Mortality: Placebo



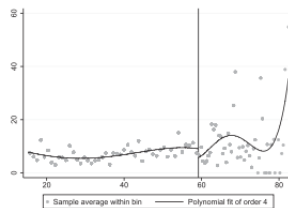
(c) Ages 5-9, all causes, post-HS



(d) Ages 25+, HS-related causes, post-HS

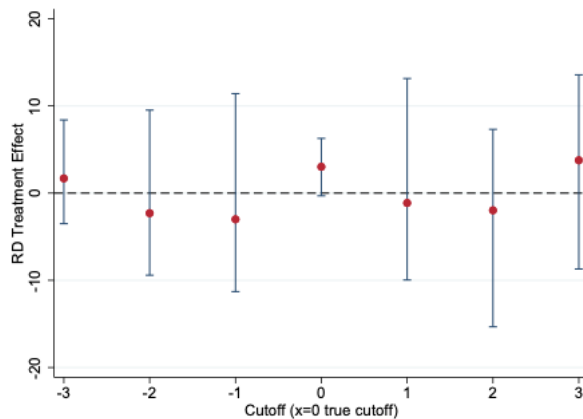


(e) Ages 5-9, injuries, post-HS



(f) Ages 5-9, HS-related causes, pre-HS

Impact of Head Start on Child Mortality: Artificial Cutoffs



Impact of Head Start on Child Mortality: RD Estimates

• Local parametric RD

$$\Rightarrow \hat{\tau}_{\{p=4, \text{full sample}\}} = -3.065, p\text{-value} = 0.005$$

$$\Rightarrow \hat{\tau}_{\{p=1, h=18\}} = -1.198, p\text{-value} = 0.071$$

$$\Rightarrow \hat{\tau}_{\{p=1, h=9\}} = -1.895, p\text{-value} = 0.055$$

• Local non-parametric RD

$$\Rightarrow \hat{\tau}_{\{p=0, \hat{h}_{MSE}=3.24\}} = -2.114, \text{robust } p\text{-value} = 0.037$$

$$\Rightarrow \hat{\tau}_{\{p=1, \hat{h}_{MSE}=6.81\}} = -2.409, \text{robust } p\text{-value} = 0.042$$

Extensions

Sharp Vs. Fuzzy RD

- Sharp RD: **score perfectly determines treatment** status
 - ⇒ All units scoring above the cutoff receive the treatment
 - ⇒ Probability of treatment jumps from 0 to 1 at c
- Fuzzy RD: **imperfect compliance** wrt treatment assignment
 - ⇒ Some units below c may be treated or vice versa
 - ⇒ Jump in probability at c may be < 1 (but > 0)

Intention-to-treat (ITT) Parameter

- Sharp RD design on the treatment assignment variable

$$\tau_{\text{ITT}} = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c]$$

- Under usual continuity assumption

$$\tau_{\text{ITT}} = \mathbb{E}[\underbrace{(Y_i(1) - Y_i(0))}_{\tau_i} (\underbrace{D_{1i} - D_{0i}}_{\substack{= 1 \text{ for compliers} \\ = -1 \text{ for defiers} \\ = 0 \text{ for always/never takers}}}) | X_i = c]$$

- ITT can be ≈ 0 even if τ is large

The Monotonicity Assumption

- Rule out the presence of **defiers**

$$P[\text{defier} | X_i = c] = 0$$

- This assumption is called **monotonicity**, since it implies that:

$$D_{1i} \geq D_{0i}, \quad \forall i$$

- $X_i \geq c$ does not decrease the probability of treatment

First Stage

- Effect of **treatment assignment on treatment status**

$$\tau_{FS} = \mathbb{E}[D_{1i} - D_{0i} | X_i = c]$$

- Under monotonicity

$$\tau_{FS} = P[D_{1i} > D_{0i} | X_i = c] = P[\text{complier} | X_i = c]$$

- First stage identifies the **proportion of compliers at the cutoff**

Recovering the ATE on Compliers (LATE)

- Instrument D_i with $\mathbb{I}(X_i \geq c)$

$$\tau_{\text{LATE}} = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c, D_{1i} > D_{0i}] = \frac{\mathbb{E}[Y_i(1) - Y_i(0) | X_i = c]}{\mathbb{E}[D_{1i} - D_{0i} | X_i = c]}$$

- Fuzzy RD parameter is “doubly local”

⇒ At the cutoff

⇒ On the subpopulation of compliers

Estimation in Fuzzy Designs

- ITT and FS are sharp RD estimators
- The FRD parameter can be estimated using [two-stage least squares](#)
- Can adapt all previous tools/packages to this case
 - ⇒ Data driven bandwidth selection
 - ⇒ Local polynomial estimation
 - ⇒ Robust bias-corrected inference

Multicutoff and Multiscore RD

1 Multiple cutoffs:

- ⇒ Cutoffs change across regions, time periods, etc
- ⇒ All units receive the same treatment when they exceed their cutoff

2 Cumulative cutoffs:

- ⇒ Treatment is multivalued
- ⇒ Different dosage of treatment depending on value of X_i

3 Multiple scores:

- ⇒ Treatment assigned based on multiple running variables

RD with Multiple Cutoffs

- Common empirical approach: **pooling**

⇒ Discrete cutoffs: $\mathcal{C} = \{c_0, c_1, \dots, c_J\}$

⇒ Re-centered running variable: $\tilde{X}_i = X_i - C_i$

⇒ Pooled estimand:

$$\tau^p = \lim_{x \downarrow 0} \mathbb{E}[Y_i | \tilde{X}_i = x] - \lim_{x \uparrow 0} \mathbb{E}[Y_i | \tilde{X}_i = x]$$

Identification under the Pooling Approach

- If the CEFs and $f_{X|C}(x|c)$ are continuous at the cutoffs, then

$$WATE = \sum_{c \in \mathcal{C}} \mathbb{E}[Y_i(1, c) - Y_i(0, c) | X_i = c, C_i = c] \omega(c)$$

- Where the **weights reflect the densities at each cutoff**

$$\omega(c) = \frac{f_{X|C}(c)P[C_i = c]}{\sum_{c \in \mathcal{C}} f_{X|C}(c)P[C_i = c]}$$

Exploiting Multiple Cutoffs

- Two drawbacks of the pooling approach:

⇒ Combines TEs for different populations

⇒ Discards variation that can identify parameters of interest

- Potential CEFs:

$$\mu_d(x, c) = \mathbb{E}[Y_i(d) | X_i = x, C_i = c], \quad d \in \{0, 1\}$$

- Cutoff-specific ATE

$$\tau(x, c) = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x, C_i = c] = \mu_1(x, c) - \mu_0(x, c)$$

RD with Cumulative Cutoffs

- Multivalued treatment $D_i \in \{d_1, d_2, \dots, d_J\}$
- Effect of switching from one dosage to the next one

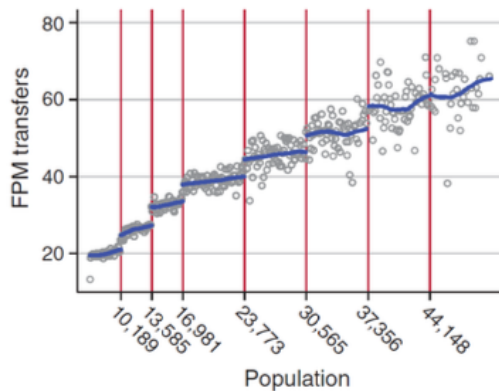
$$\tau_j = \mathbb{E}[Y_i(d_j) - Y_i(d_{j-1}) | X = c_j]$$

- Under continuity assumption

$$\tau_j = \lim_{x \downarrow c_j} \mathbb{E}[Y_i | X_i = x] - \lim_{x \uparrow c_j} \mathbb{E}[Y_i | X_i = x]$$

RD with Cumulative Cutoffs: Example

- Effect of federal transfers to municipalities on corruption in Brazil
- Federal transfers change discontinuously at population thresholds



RD with Multiple Scores (Boundary RD)

- Bivariate score: $\mathbf{X}_i = (X_{1i}, X_{2i})$
- Treat/control regions $\mathcal{A}_1, \mathcal{A}_0$ and **boundary** $\mathcal{B} = \text{bd}(\mathcal{A}_1) \cap \text{bd}(\mathcal{A}_0)$
- **Multidimensional RD** parameter:

$$\tau(\mathbf{b}) = \mathbb{E}[Y_i(1) - Y_i(0) | \mathbf{X}_i = \mathbf{b}], \quad \mathbf{b} \in \mathcal{B}$$

- ATE at each point in the boundary $\mathcal{B}$

RD with Multiple Scores (Boundary RD)

- Estimating a whole curve of treatment effects may not be feasible
- Define **distance measure**, e.g. $d(\mathbf{X}_i, \mathbf{b}) = \sqrt{(X_{1i} - b_1)^2 + (X_{2i} - b_2)^2}$
- Run a pooled RD on **(unidimensional) running variable** $\tilde{d}_i$
- Define a set of boundary points of interest and estimate an RD at each of the specified points
- Helps explore heterogeneity in treatment effects across boundary points

School Choice Models

Revealed Preference Analysis of School Choice Models

- Unlike traditional consumer settings, education markets do not use prices as clearing mechanism
 - ⇒ Empirical approaches often rely on the systems for rationing or admitting students to schools/colleges
- Revealed preference approach
 - ⇒ Use data from decisions and assignments to estimate the distribution of preferences over schools
- Leverage estimated preferences to generate counterfactual student-school equilibria

School Choice Mechanisms

- R_{ik} is student i 's submitted ranked-order list (RoL), where k is the position of the ranked school
- s_{ij} is student i 's priority in school j
- q_j is a vector of schools' capacities
- A **mechanism** ϕ maps the tuple (R, s) into school assignments μ

$$\mu_{ij} = \begin{cases} 1 & \text{if student } i \text{ is assigned to school } j \\ 0 & \text{otherwise} \end{cases}$$

- **Feasible matching:** $\sum_i \mu_{ij} \leq q_j, \forall j$

The Deferred Acceptance (DA) Mechanism

- Each student applies to the highest ranked school
 - ⇒ Schools **tentatively hold** students with highest priority up to q_j , remaining students are rejected
- Previously rejected students apply to their highest ranked school that has not yet rejected them
 - ⇒ **Previously held** students are considered along with the **new applicants**
 - ⇒ Schools tentatively hold the highest priority students while the remaining students are rejected
- The algorithm terminates whenever
 - ⇒ No students are rejected
 - ⇒ All school slots q_j are filled
 - ⇒ All rejected students have applied to each of their ranked schools

DA Properties

- When priority indices are not school-specific, DA is equivalent to **serial dictatorship**
 - ⇒ Students, in the order of their priority indices, are allowed to choose among the remaining schools

- DA generates an allocation determined by a vector of cutoffs c_j and priorities s_{ij} such that

$$\Omega_i = \{j : s_{ij} > c_j\}$$

- Each student is **assigned to the highest ranked school in her choice set**
 - ⇒ There is no pair (i, j) : (i) $i \succ j$ over μ_{ik} and (ii) if school does not have empty slots, then $s_{ij} > c_j$
- Submitting **RoLs that coincide with ordinal preferences** is a weakly dominant strategy

Stable Matching

- Stability of the matching equilibrium is more likely when
 - ⇒ Students know their (ex-post) scores s_{ij}
 - ⇒ School cutoffs c_j are predictable by both students and schools (large matching market)
- If student i is assigned to school j , then $u_{ij} > u_{ik} \quad \forall k \in \Omega_i$
- Stability of the final assignment allows us to infer preferences with data on matching outcomes only
 - ⇒ Or where submitted RoLs may not be truthful

Estimating Preferences Under Stable Matching

- Stability defines a **discrete choice model with personalized choice sets**

⇒ E.g., $u_{ij} = \tilde{u}_{ij}(x_i, z_j; \theta) + \epsilon_{ij}$, with $\epsilon_{ij} \sim \text{T1EV}$, then choice probabilities are

$$P(\mu_{ij} = 1 | x_i, z_j; \theta) = \frac{\exp(\tilde{u}_{ij})}{\sum_{k \in \Omega_i} \exp(\tilde{u}_{ik})}$$

⇒ And the log likelihood is

$$LL(\theta) = \sum_{i=1}^N \sum_{j=1}^J \log \left(\frac{\exp(\tilde{u}_{ij})}{\sum_{k \in \Omega_i} \exp(\tilde{u}_{ik})} \right)$$

- Assignments provide no information about **preferences for schools that are not in Ω_i**

⇒ Requires extrapolation from students with larger choice sets to those with smaller choice sets

$$\epsilon_{ij} \perp s_j | x_i, z_j$$

Normalizations in Discrete Choice Models

- Only **differences in utility** are identified
 - ⇒ We can add an arbitrary number to both sides of the inequality
 - ⇒ E.g., we assume $u_{i0} = 0$ (outside option) and parameters identify differences with $j = 0$
- The **scale of utility is irrelevant**
 - ⇒ Also multiplying both sides by something > 0 doesn't affect the inequality
 - ⇒ We either normalize $\text{Var}(\epsilon_{ij})$, or divide all parameters by a money metric (e.g. a price coefficient)

IIA and Unobserved Heterogeneity

- Unobserved heterogeneity enters only through the iid type 1 extreme value on ϵ_{ij}
 - ⇒ No unobserved heterogeneity in valuation of school characteristics z_j
 - ⇒ No correlation of unobserved tastes over different alternatives
- This implies the **Independence of Irrelevant Alternatives** (IIA) property
 - ⇒ The ratio of two probabilities does not depend on a third alternative

$$\frac{P(\mu_{ij} = 1 | x_i, z_j)}{P(\mu_{ik} = 1 | x_i, z_j)} = \exp(u_{ij} - u_{ik})$$

- ⇒ Any model with iid and full-support shocks will produce similar patterns (e.g. multinomial probit)

Random Coefficients Logit (aka Mixed Logit)

- One solution to IIA is to allow for correlation in ϵ_{ij} within a “nest” (Nested Logit)
- A popular alternative is to allow for **unobserved heterogeneity in taste parameters**: $\beta_i z_j$
 - ⇒ E.g. Normal random coefficients: $\beta_i \sim N(x_i' \gamma, \Sigma)$
 - ⇒ Finite mixture: $\beta_i \in \{\beta_1, \dots, \beta_k\}$ and $P(\beta_i = \beta_k) = \omega_k$
- Choice probabilities have **no closed form solution**

$$P(\mu_{ij} = 1 | x_i, z_j; \theta) = \int \frac{\exp(\tilde{u}_{ij})}{\sum_{k \in \Omega_i} \exp(\tilde{u}_{ik})} f(\beta; \theta) d\beta$$

Substitution Patterns Under Mixed Logit

- The ratio of mixed logit probabilities, P_{ij}/P_{im} , depends on **all the data** ($z_k, \forall k \neq j, m$)
 $\Rightarrow$ Denominators of the logit ratio are inside the integrals and therefore do not cancel

- Elasticity of choice probabilities is **different for each alternative j**

$$E_{P_{ij}, z_m} = -z_m \int \theta_z \frac{\exp(\tilde{u}_{im}(\theta))}{\sum_{k \in \Omega_i} \exp(\tilde{u}_{ik}(\theta))} \left[\frac{\exp(\tilde{u}_{ij}(\theta))}{\sum_{k \in \Omega_i} \exp(\tilde{u}_{ik}(\theta))} / P_{ij} \right] f(\beta; \theta) d\beta$$

- Substitution pattern depends on the correlation between logit ratios of m and j over values of β
- **Any random utility model can be approximated by a mixed logit**

Estimation: Simulated Maximum Likelihood (SMLE)

- Draw many values β^r from $f(\beta|\theta)$ and calculate the logit ratio with each draw

$$\tilde{P}_{ij} = \frac{1}{R} \sum_{r=1}^R \frac{\exp(\tilde{u}_{ij}(\beta^r))}{\sum_{k \in \Omega_i} \exp(\tilde{u}_{ik}(\beta^r))}$$

- The simulated probabilities are inserted into the log-likelihood function

$$SLL(\theta) = \sum_{i=1}^N \sum_{j=1}^J \log(\tilde{P}_{ij})$$

- SMLE is efficient but not consistent because simulation error enters non-linearly through the log

Teacher Compensation and Structural Inequality

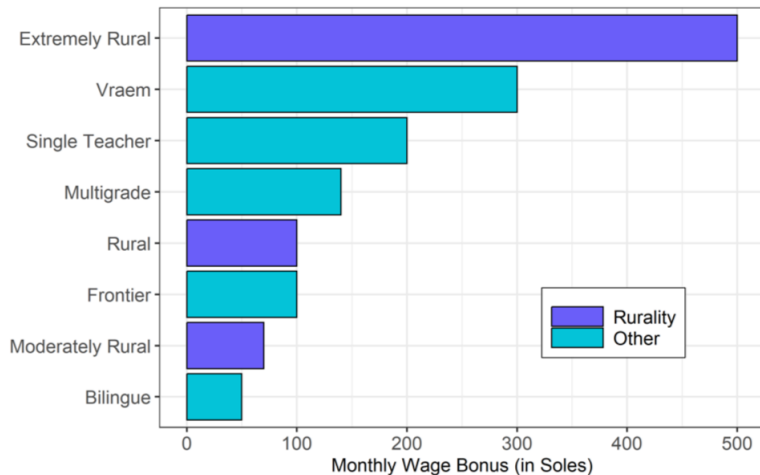
Bobba-Ederer-Leon-Neilson-Nieddu (JPE, forthcoming)

- ① Descriptives on nation-wide allocation of public teachers in Peru
 - ⇒ Document **large urban-rural gaps** in teacher quality and student test scores
- ② RD-based evidence of teacher **wage bonuses in rural locations**
 - ⇒ Teacher competency $\uparrow$ by 0.42σ + student test scores $\uparrow$ by $0.21-0.27\sigma$
- ③ Model of teacher school choice/value added to study **aggregate policy effects**
 - ⇒ Equity and efficiency gains of teacher reallocation vs. recruitment

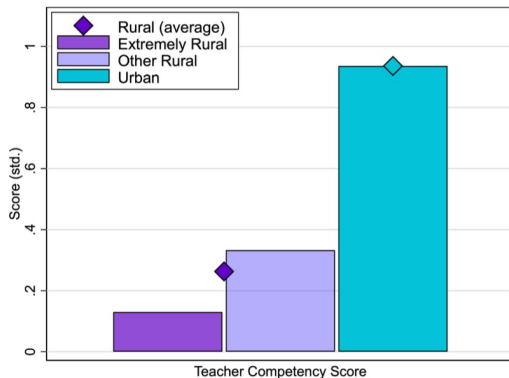
Teachers in Peru

- Focus on **contract teachers in public primary schools** for coverage in rural areas
- Serial dictatorship algorithm with priorities determined by competency score
- Wages are fixed up to **school-specific bonuses**
- We can link teachers to student-level test scores

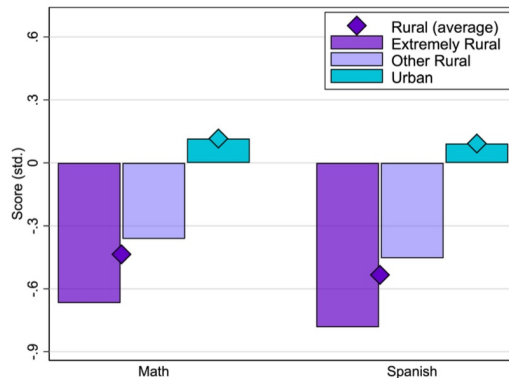
Wage Bonuses for Teachers in Disadvantaged Schools



Inequality of Education Inputs and Output

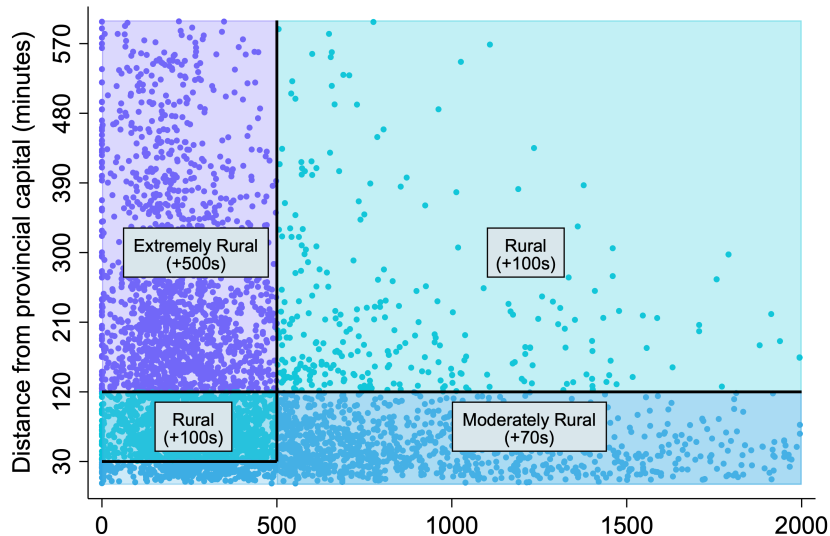


a) Teacher Competency by Urban/Rural

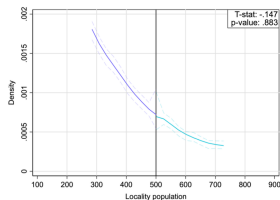


b) Student Achievement by Urban/Rural

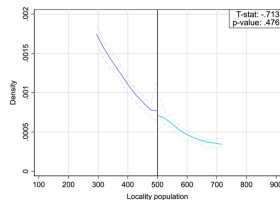
The Rural Wage Bonus Policy



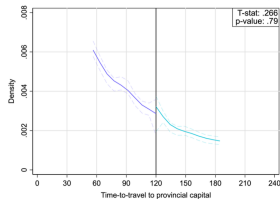
Density Tests Around Extremely Rural Cutoff



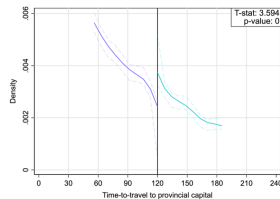
a. Population (2016)



b. Population (2018)



c. Time-to-travel (2016)



d. Time-to-travel (2018)

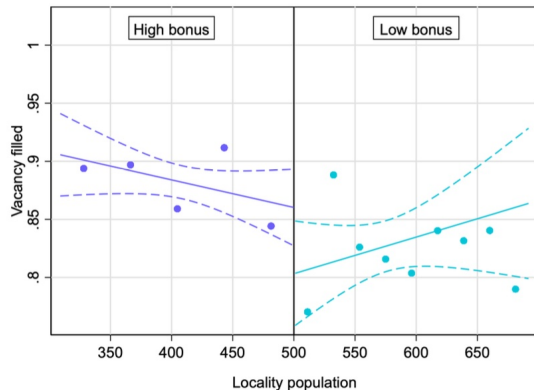
Sharp RD Along Population Cutoff

- We rely on **population-based assignment** rule for rural schools with $\text{dist} > 30\text{min}$
- Weighted average increase in wages of 11%
- Given continuity of potential outcomes around the population cutoff

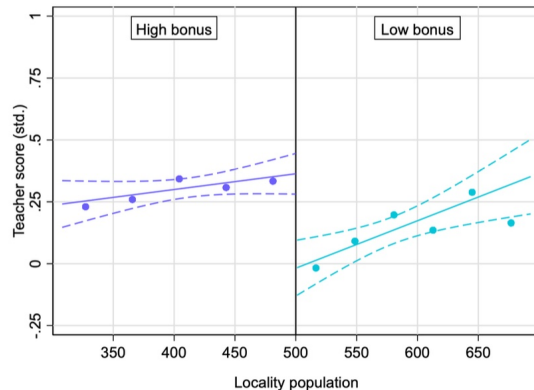
$$y_{ijt} = \gamma_0 + \gamma_1 \mathbf{1}(\text{pop}_{jt} < \text{pop}_c) + g(\text{pop}_{jt}, \text{pop}_c) + \delta_t + u_{ijt}$$

- $g(\cdot)$: flexible polynomial on population of the locality of school j
 - δ_t : indicator for year of assignment
 - u_{ijt} : error term, clustered at the school-year level
- Estimate γ_1 **non-parametrically within MSE-optimal** bandwidths

Rural Bonus and Teacher Choices over Job Postings



a) Vacancy Filled



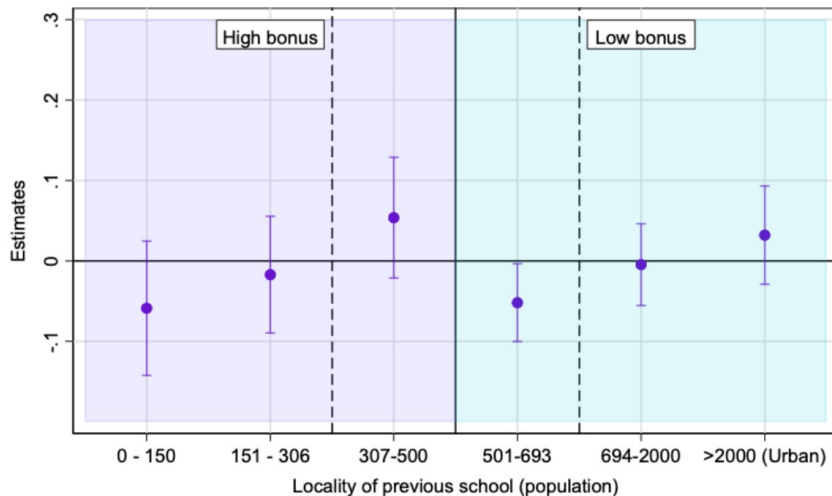
b) Competency Score

Rural Bonus and Teacher Choices over Job Postings

	(1) Vacancy filled	(2) Preferences	(3) Teacher Score (Std.)
High Bonus	0.063 (0.048)	0.108 (0.031)	0.418 (0.106)
Bandwidth	190.896	162.270	193.380
Schools	1478	1157	1388
Observations	3524	2575	3068

NOTES. This table reports the effect of crossing the population threshold on several outcomes. In Column (1), the outcome variable is an indicator for whether the vacancy is filled in the national recruitment drive. Column (2) reports the effect on the rank at which a vacancy is chosen within a school district, normalized to range from zero to one. Column (3) reports the results on the competency score obtained by teachers in the centralized test. In all regressions, we use our choice sample and exclude urban schools (i.e., those located in areas with more than 2,000 inhabitants) and schools situated within 30 minutes of the nearest provincial capital. In Columns (2) and (3), the sample is further restricted to vacancies filled in the national recruitment drive. Cells report bias-corrected regression-discontinuity estimates obtained using the robust estimator proposed by [Calonico et al. \(2014\)](#). Regressions are estimated within a mean-squared-error-optimal bandwidth, reported at the bottom of the table. Standard errors are clustered at the school×year level.

Rural Bonus and the Origin of Newly Recruited Teachers



Rural Bonus and Student Achievement

Panel A: Dependent Variable is Spanish Test (z-score)

	(1)	(2)	(3)
	New teacher	No vacancy	All
High Bonus	0.340 (0.174)	-0.080 (0.124)	0.206 (0.084)
Bandwidth	92.657	143.045	114.510
Schools	221	436	931
Observations	3830	6578	18828

Panel B: Dependent Variable is Math Test (z-score)

	(1)	(2)	(3)
	New teacher	No vacancy	All
High Bonus	0.543 (0.221)	-0.040 (0.143)	0.265 (0.099)
Bandwidth	79.168	144.756	110.039
Schools	192	442	889
Observations	3408	6699	18083

What is the Rationale for the Model?

- The RD evidence is limited to the local effect of the rural wage bonus
- What is the **effect of the policy** on urban-rural gaps in education outcomes?
- Can we characterize **more effective teacher-school allocations**?
- Can we achieve those with **alternative wage schedules**?

An Empirical Matching Model of Teachers and Schools

- ① A school choice framework recovers **teachers' preferences**
 - ⇒ Flexibly parametrize preference distributions over wages and non-wage amenities
- ② A Teacher Value-Added (TVA) model that maps teacher sorting into **student achievement**
 - ⇒ $TVA = ATE + \text{Match Effect (students' prior achievement and demographics)}$
 - ⇒ TVA parameters correlate with teachers' preferences

Teacher Preferences

- Teacher i utility from school j (off-platform $j = J + 1$ + outside option $j = 0$):

$$u_{ijt} = \alpha_i w_{jt} + x_i' \Theta_1 z_{jt} + x_i' \Theta_2 d_{ijt} + \kappa_{c(j)} + \eta_{ijt}$$

$$u_{iJ+1t} = \gamma_{0t} + x_i' \gamma_1 + \eta_{iJ+1t}$$

$$u_{i0t} = \beta_{0t} + \beta_i + \eta_{i0t}$$

- Correlated Random coefficients: $\theta_i = (\log \alpha_i, \beta_i)$, $\theta_i | x_i \sim \mathcal{N}(\mu'_\theta x_i, \Sigma_\theta)$
- Serial dictatorship $\Rightarrow$ discrete choice model with observed choice sets

$$D_{it} = \arg \max_{j \in \Omega(s_{it}) \cup \{0, J+1\}} u_{ijt}$$

$\Rightarrow$ Two choices per teacher + variation in choice sets within and across teachers

Teacher Value Added

- Student ℓ **potential outcome** when matched with teacher i :

$$Y_{\ell i} = + \underbrace{\delta_{0i}}_{\text{teacher ATE}} + \underbrace{X'_{\ell 1} \delta_{1i}}_{\text{Match Effects}} + \underbrace{X'_{\ell 2} \delta_2}_{\text{School/classroom effect}} + \epsilon_{\ell i}$$

- Correct for **selection on teachers' unobserved preference** for assigned schools

$$\mathbb{E}[Y_{\ell} | X_{\ell 1}, X_{\ell 2}, S_{\ell} = i, \eta_{ij(\ell)}] = \delta_{0i} + X'_{\ell 1} \delta_{1i} + X'_{\ell 2} \delta_2 + \varphi \times (\eta_{ij(\ell)} - \mu_{\eta})$$

$\Rightarrow \varphi$ captures selection on unobserved teacher-school match effects

$\Rightarrow d_{ij(\ell)}$ and $w_{j(\ell)}$ are **excluded from students' potential outcomes**

Connecting Teacher Preferences and Value Added

- We allow TVA, $\delta_i = (\delta_{0i}, \delta_{1i})$, to correlate with teacher preferences over schools

$$(\theta_i, \delta_i) | x_i \sim \mathcal{N} \left[\begin{pmatrix} \mu'_\theta x_i \\ \mu'_\delta x_i \end{pmatrix}, \begin{pmatrix} \Sigma_{\theta, \theta} & \Sigma_{\theta, \delta} \\ \Sigma_{\delta, \theta} & \Sigma_{\delta, \delta} \end{pmatrix} \right]$$

⇒ $\Sigma_{\theta, \theta}$: how teachers value **non-wage benefits/outside option** against wages

⇒ $\Sigma_{\theta, \delta}$: **selection on teacher quality** resulting from changes in wages

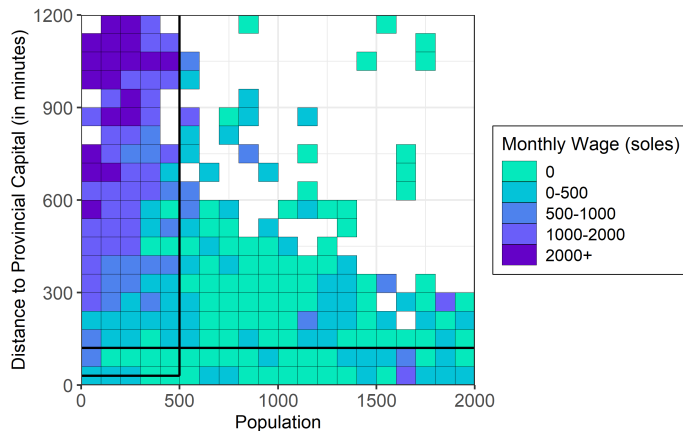
⇒ Given $\hat{\delta}_i$, we can identify and estimate the distribution of θ_i conditional on δ_i , which pins down $\Sigma_{\theta, \delta}$

Willingness to Pay for Non-Pecuniary Job Attributes

	Bottom Decile (1)	Median (2)	Top Decile (3)
Amenity Index	0.028	0.114	0.450
Multigrade	-1.880	-0.435	-0.075
Single Teacher	-4.263	-1.031	-0.193
Bilingual	-3.270	-0.747	0.338
Share of Female Students	-0.790	-0.103	0.156
Share of Quechua Students	-4.011	-0.898	1.003
Average Math Score	-0.293	0.001	0.236
Average Spanish Score	-0.018	0.015	0.110
Distance Spline: 0-20km	-0.451	-0.124	-0.034
Distance Spline: 20-100km	-0.232	-0.065	-0.018
Distance Spline: 100km+	-0.039	-0.010	-0.003

- Non-wage attributes induce **vertical+horizontal differentiation** across schools

Rural vs. Urban Non-Pecuniary Utility Differences



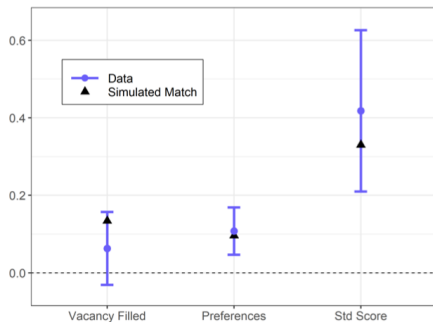
- Utility differences are **merely compensated by the wage bonus** policy

Distribution of Value-Added Parameters

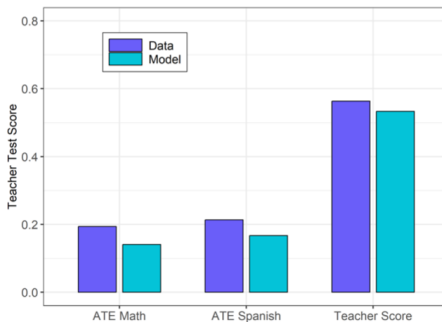
	Mean Baseline	SD	Correlation		Covariance RC	
			Lagged Score	Female	Wage	Outside Option
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Math test scores</i>						
ATE	0	0.384 (0.006)	0.078 (0.031)	0.252 (0.082)	-0.049 (0.019)	-0.089 (0.046)
Lagged Score	0.498 (0.011)	0.150 (0.004)		0.021 (0.082)	-0.019 (0.011)	0.001 (0.023)
Female	-0.072 (0.014)	0.097 (0.011)			-0.008 (0.015)	-0.006 (0.030)
Unobserved Preferences (φ)	-0.007 (0.006)	-	-	-	-	-

- Moving one SD up the distribution of ATE implies an increase in student math score of 0.38σ
- Large gains from attracting better teachers along **both absolute and comparative advantages**

Model Fit



a) Threshold-Crossing Effects



b) Urban-Rural Inequality

- Estimated **model replicates the RD evidence** induced by the rural wage bonus
- Good fit on moments away from the population threshold (urban-rural gaps, etc.)

Counterfactual 1: Gains from Teacher Reallocation Vs. Recruitment

- Predict teachers' choices over schools with and without rural bonus
 - ⇒ Simulate u_{ijt} from estimated parameters and a random draw of ϵ_{ijt} and θ_i
 - ⇒ Compute the stable matching equilibrium using the teacher-proposing DA algorithm
- Simulate largest possible gains by reallocating the existing pool of public teachers
- Contrast this benchmark with changes in the composition of the public-teacher pool

Teacher Sorting and Inequality

	Status quo (1)	Value-added Maximization (2)
<i>Panel A: ATE Math</i>		
Urban	0.112	0.053
Rural	-0.029	0.050
<i>Other Rural</i>	0.001	0.056
<i>Extremely Rural</i>	-0.085	0.038
Total	0.052	0.052
<i>Panel B: Match Effects Math</i>		
Urban	0.004	0.044
Rural	0.003	0.091
<i>Other Rural</i>	0.002	0.075
<i>Extremely Rural</i>	0.005	0.122
Total	0.003	0.064

- Gains from reallocation in rural schools by 0.17σ ($=0.08\sigma$ in ATE + 0.09σ in Match Effects)
- Gains from the extensive margin of recruiting exceed those from match effects ($0.09\sigma > 0.06\sigma$)

Counterfactual 2: Compensation Policy

- Y_{ij} is teacher i 's predicted value-added at school j (ATE + match effects)
- For $w = (w_1, \dots, w_J)$, let $D(w) = \{D_i(w)\}_{i=1}^N$ denote the associated teacher-school assignment

$$\max_{\{i:D_i(w)=j\}} Y_{ij} \geq \bar{c}, \quad \forall j \in \mathcal{S} \quad (C1)$$

$$D(w) \text{ is stable given } w \text{ and school priorities } Y_{ij} \quad (C2)$$

- We construct the counterfactual schedule through **iterative wage adjustments**
 - $\Rightarrow$ Each targeted school $j \in \mathcal{S}$ bids wage upward until (C1) is satisfied
 - $\Rightarrow$ Recomputing the assignment accounts for the equilibrium effects on teacher sorting across all schools

Counterfactual Teacher Compensation Policy

	Status Quo (1)	Counterfactual Policy (2)
<i>Panel A: ATE Math</i>		
Urban	0.112	0.143
Rural	-0.029	0.082
<i>Other Rural</i>	0.001	0.091
<i>Extremely Rural</i>	-0.085	0.059
Total	0.052	0.119
<i>Panel B: Match Effects Math</i>		
Urban	0.004	0.009
Rural	0.003	0.019
<i>Other Rural</i>	0.002	0.011
<i>Extremely Rural</i>	0.005	0.040
Total	0.003	0.013
<i>Panel C: Monthly Wage Bonus</i>		
Urban	0	0
Rural	235	227
<i>Other Rural</i>	86	152
<i>Extremely Rural</i>	500	367
Total	122	120

- TVA increases by 0.13σ in rural schools and by 0.04σ in urban schools (mostly [through ATE gains](#))

RDD $\Leftrightarrow$ Structural Model

- RD evidence is local in nature and may neglect indirect/equilibrium effects
- The model is a tool to extrapolate the policy effect away from the RD cutoff
 - $\Rightarrow$ Quantify aggregate effect of the policy reform
 - $\Rightarrow$ Characterize gains from alternative teacher assignments
 - $\Rightarrow$ Design alternative teacher compensation schemes
- RD variation is embedded in the model, enhancing its credibility